

11/4/25

Defn of topological space is compact if whenever $(x_i)_{i \in I}$ is an indexed family in X and $\mathcal{U}$ is an ~~ult-filter~~ ^{filter} on I , then there exists $l \in X$ st $x_i \xrightarrow{\mathcal{U}} l$

Defn 2. Adj. to above, but replace "ult-filter" with "filter"

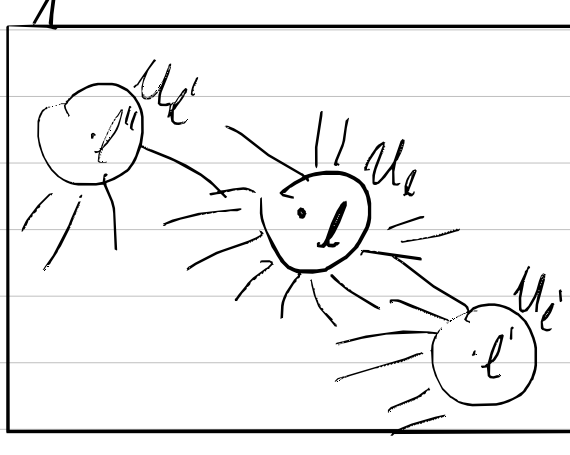
Def 2 $\Rightarrow$ Def 1 is trivial as all ult-filters are filters

Now showing Def 1 $\Rightarrow$ Def 2.

Fix $(X, \mathcal{C}(x))_{x \in X}$ as above and fix an indexed family $(x_i)_{i \in I}$ in X .
Now fix a filter $\mathcal{F}$ on I .

We must show that there exists $l \in X$ st

$$x_i \xrightarrow{\mathcal{F}} l$$



Assume this is not the case, so for every $l \in X$ and for U_l of l st $\exists i \in I : x_i \notin U_l \nexists \mathcal{F}$.

Consider sets

$$U_l^c \text{ and } l \text{ vary}$$

These sets have the f.p. so we find an ult-filter $\mathcal{U}$ that contains all the sets
by construction there is no $l \in X$ st $x_i \xrightarrow{\mathcal{U}} l$

This contradicts Def 1. $\square$

Exercise.

Recall a topological space $(X, \mathcal{C}(x))_{x \in X}$ is Hausdorff if, whenever $x, y \in X$ are distinct, then there exists $U \in \mathcal{C}(x)$ and $V \in \mathcal{C}(y)$ st $U \cap V = \emptyset$

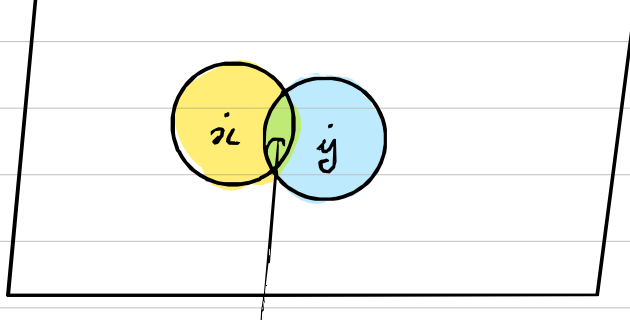
prove that if every limit is unique then the space is Hausdorff.

Suppose $(X, \mathcal{C}(x))_{x \in X}$ satisfies the following

Of $(x_i)_{i \in I}$ and $\mathcal{F}$ is a filter on I
 $x_i \xrightarrow{\mathcal{F}} l$
 $x_i \xrightarrow{\mathcal{F}} l'$

implies $l = l'$

On proving the above, assume there is not Hausdorff, i.e. there exists $x, y \in X$ st $U \in \mathcal{C}(x)$ and $V \in \mathcal{C}(y)$ st $U \cap V \neq \emptyset$, so



show $U \cap V$ has f.p.

New Chapter

Recall:

a graph (G, R) has
 G : a set of nodes
and R : a relation

Now consider a family of graphs $(G_n, R_n)_{n \in \mathbb{N}}$ $\xrightarrow{\mathcal{U}}$ (G, R)
Graph limit (Graphons)

(G, R) is the $\mathcal{U}$ -limit of the family $(G_n, R_n)_{n \in \mathbb{N}}$.

Suppose we are given a family $(G_1, R_1), (G_2, R_2), \dots$ of graphs and an ult-filter $\mathcal{U}$ on $\mathbb{N}$. We define the $\mathcal{U}$ -limit (G, R) of the sequence $(G_n, R_n)_{n \in \mathbb{N}}$ as follows,

Consider the Cartesian product

$$\prod_{n \in \mathbb{N}} G_n$$

which consists of all sequences $(a_n)_{n \in \mathbb{N}}$, where $a_n \in G_n \forall n \in \mathbb{N}$

then $a \in G$,

per R
Suppose $a, b \in \prod_{n \in \mathbb{N}} G_n$

$$a = (a_1, a_2, \dots)$$

$$b = (b_1, b_2, \dots)$$

Def $a R b = \{n \in \mathbb{N} : a_n R_n b_n\} \in \mathcal{U}$
 $a \not R b = \{n \in \mathbb{N} : a_n R_n b_n\} \notin \mathcal{U}$

This is taking a majority, eg. (a.e.)

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Graph Limits - Transfer Theorem
Suppose we are given a sequence of graphs

$$\begin{aligned} G_1 &= (G_1, R_1), \\ G_2 &= (G_2, R_2), \\ &\vdots \end{aligned}$$

Q1: Do these graphs converge to a graph-limit.

Q2: Which properties of $G_1, G_2, \dots$ "transfer" to the graph limit and which do not?

Fix an ultra-filter $\mathcal{U}$ on the indexed set $\mathbb{N} = \{1, 2, \dots\}$
we will show there is a graph G st

$$G_n = (G_n, R_n)_{n \in \mathbb{N}} \xrightarrow{\mathcal{U}} G = (G, R)$$

Let's assume $\mathcal{U}$ is not principal

Remark, given any sets $A_1, A_2, \dots$ define the Cartesian product

$$\prod_{n \in \mathbb{N}} A_n = \{(x_n)_{n \in \mathbb{N}} : x_n \in A_n\}$$

*This should start the chapter *

Note: (x_n) is the n -th coordinate

Of $x = (x_n)_{n \in \mathbb{N}}$ is a sequence in $\prod A_n$
this is called the n -th coordinate of x .

The function

$x \mapsto x_n$
is called the n -th projection

$$p_n : \prod_{j \in \mathbb{N}} A_j \xrightarrow{x \mapsto x_n} A_n$$

Construction of $\mathcal{U}$ -limit of $G_1, G_2, \dots$

1st

$$G = \prod_{n \in \mathbb{N}} G_n$$

Fix $x, y \in G$, then

$$x = (x_1, x_2, \dots)$$

$$y = (y_1, y_2, \dots)$$

Def $x \sim y$

if $\{n : x_n = y_n\} \in \mathcal{U}$

Claim: this is an equivalence relation

WTS:

Reflexive $\{n : x_n = x_n\} = \mathbb{N} \in \mathcal{U}$

Symmetric if $x \sim y$ then

$$\{n : x_n = y_n\} = \{n : y_n = x_n\} \in \mathcal{U}$$

Transitive $x \sim y, y \sim z \Rightarrow x \sim z$

Assume $x \sim y, y \sim z$

$$\{n : x_n = y_n\} \in \mathcal{U}$$

$$\{n : y_n = z_n\} \in \mathcal{U}$$

$$\text{then } x \sim z = \{n : x_n = z_n\} \supseteq \{n : x_n = y_n \text{ or } y_n = z_n\} \in \mathcal{U}$$

Of $x, y \in G$, regard x, y as equal if $x \sim y$

2nd

For $x, y \in G$ define

$$x R y \text{ iff } x_n R_n y_n \text{ (a.e.)}$$

$$\text{ie } x R y = \{n : x_n R_n y_n\} \in \mathcal{U}$$

$$\text{and } x R y = \{n : x_n R_n y_n\} \in \mathcal{U}$$

The graph $G_{\mathcal{U}} = (G, R)$ is called the $\mathcal{U}$ -ultraproduct or the $\mathcal{U}$ -limit of $G_1, G_2, \dots$

Remark: The preceding construction works just as well if we replace $\mathbb{N}$ by any inf. set I or if we replace G_n by relational structure $(G_i, R_i)_{i \in I}$

Notation

$$G_n \xrightarrow{\mathcal{U}} G_{\mathcal{U}}$$

$$G_{\mathcal{U}} = \mathcal{U}\text{-lim } G_n$$

$$G_{\mathcal{U}} = \prod_{n \in \mathbb{N}} G_n / \mathcal{U}$$

Transfer Theorems

Def 1st order statements are made up of connectives and quantifiers

Let $\mathcal{U}$ be an ultrafilter on I an indexed set

Then if $A, B \subseteq I$

$$1) A \cap B \in \mathcal{U} \text{ iff } A \in \mathcal{U} \text{ and } B \in \mathcal{U}$$

$$2) A \cup B \in \mathcal{U} \text{ iff } A \in \mathcal{U} \text{ or } B \in \mathcal{U}$$

$$3) A^c \in \mathcal{U} \text{ iff } A \notin \mathcal{U}$$

Pf. 1.

$\Rightarrow$ This follows from $A \supseteq A \cap B$ and $B \supseteq A \cap B$

so $A \in \mathcal{U}$ and $B \in \mathcal{U}$

$\Leftarrow$ By def of a filter $A \cap B \in \mathcal{U}$.

2.

$\Rightarrow$ To prove this assume $A \cap B \in \mathcal{U}$

Then $A \in \mathcal{U}$ or $B \in \mathcal{U}$ otherwise

$$A^c, B^c \in \mathcal{U}, \text{ then } A^c \cap B^c \in \mathcal{U}, \text{ by De Morgan } (A \cup B)^c = A^c \cap B^c \in \mathcal{U}$$

$\Leftarrow$ The reverse follows from choice under super sets

3.

by def of Ultrafilter

Corollary Consider a family of graphs

$$G_1 = (G_1, R_1)$$

$$G_2 = (G_2, R_2)$$

Let G be the $\mathcal{U}$ -lim of $G_1, G_2, \dots$

if $p(x), q(x)$ are properties of graphs,

we say $p(x)$ holds in G iff

$$\exists x \in G : p(x) \text{ true } \exists \in \mathcal{U}$$

similar for $p(x) \wedge q(x), p(x) \vee q(x), \neg p(x)$ and $\exists, \forall$.